

The Common Aero Vehicle: Space Delivery System Of The Future

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Abstract. The Common Aero Vehicle (CAV) is a concept which describes a space reentry aeroshell launched into space on a suitable vehicle, which then survives atmospheric reentry, reduces its speed to low Mach numbers or even sub-Mach, and dispenses a cargo, payload or weapon in the Earth's atmosphere. The conceptual CAV might be propelled into space by any of a number of present or future launch platforms and dispense a wide variety of cargoes, payloads or weapons. Development of the CAV capability will satisfy future requirements enunciated in numerous national visions, future studies, and military plans and will ultimately be necessary to fully realize the opportunities inherent in operating from, through, and in space and give true meaning to the phrases global reach, global power projection, and global engagement. This paper describes notional CAV characteristics, identifies future missions, enumerates issues regarding launch platforms, cargo, payloads and weapons, disposition of the CAV upon mission completion, and identifies studies and plans which advocate a CAV capability in the future.

Introduction

In the Fall of 1996, Air Force Space Command (AFSPC) assembled over one hundred military and civilian experts and initiated a year-long study regarding the feasibility of a Military Space Plane (MSP) (since redesignated the Space Operations Vehicle or SOV). These space transportation experts constituted an overarching Integrated Process Team (IPT) which explored, among many other aspects of the MSP, the missions which might be accomplished with such a reusable space-launch system. Early in the studies it was realized that a MSP could support missions not only in space, but also more conventional missions with *near-real-time to real-time* delivery of cargo, payloads, or weapons anywhere in the Earth's atmosphere or on the Earth's surface. Support of atmospheric or surface missions, however, required some form of protection for the cargoes, payloads or weapons returning at extremely high speeds into the Earth's atmosphere. Given the daunting distance and speed problems associated with realization of such delivery capability from space, some form of space launch with subsequent reentry to or through the atmosphere is the only practical means of achieving near-real-time global reach in the foreseeable future. As part of the overarching MSP IPT, Air Force Space Command began investigating concepts for the CAV, a reentry vehicle which could depart a spacecraft and independently return, to accomplish missions in the Earth's atmosphere.

Early Work of Note

Early CAV investigations quickly made it apparent that considerable work had previously been accomplished on the space reentry problem. Present CAV concepts continue a long line of previous maneuverable reentry vehicle studies and technology programs stretching back almost 40 years. Work began in the early 1960s on hypersonic maneuvering reentry studies which took several conceptual tracks and ultimately contributed to a number of later space programs ranging from manned space flight to advanced maneuvering reentry concepts for the nation's ICBM forces.

Under the Air Force Spacecraft Technology and Advanced Reentry Tests (START) Research Program, numerous concepts were developed, designed, and flown in an attempt to produce a reusable, maneuverable, reentry vehicle capable of being flown from Earth orbit to a precise landing point on earth. The first phase of the START Program, called the Aerothermodynamic/elastic Structural Systems Environmental Tests (ASSET) program, designed and tested lifting bodies built by McDonnell Aircraft Corporation which derived aerodynamic lift from their shape alone. During the mid-1960s, six ASSET vehicles, launched on Thor boosters, measured the hypersonic environment and tested materials and structural design concepts related to the maneuvering reentry problem. In a second part of the START Program, called Precision Recovery Including Maneuvering Entry (PRIME), advanced lifting body vehicles built by Martin-Marietta were launched on Atlas missiles to further

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explore the development of manned and unmanned lifting body vehicles.¹

Other companies, including Rockwell, Boeing, Martin-Marietta, Lockheed, TRW, and General Electric, supported the military in technology programs designed to develop and test space reentry vehicle technologies. In the mid-1960s, Boeing participated in maneuvering control and ablation technology studies, early flight tests, and design of a maneuvering Boost Glide Reentry Vehicle concept. This work was followed by operational maneuvering reentry vehicle and maneuvering decoy studies which produced an advanced reentry vehicle design culminating in maneuvering reentry vehicle control experiment flight tests in the early 1970s.²

Through the 1970s, Boeing continued work on reentry vehicles, designing a Small Evader Vehicle, a Precision Guided Reentry Vehicle, and flight testing an Advanced Maneuvering Reentry Vehicle. In the 1980s, Boeing conducted design studies under their Maneuvering Technology Program, designing a Small High Accuracy Reentry Vehicle, a Defense Suppression Weapon/Vehicle, and conducting a Technology Demonstration of their Maneuvering Reentry Vehicle. During this time TRW was also developing concepts and designs for Maneuvering Reentry Vehicles (MaRV), Advanced Maneuvering Reentry Vehicle (AMaRV) and the High Performance Maneuvering Reentry Vehicle (HpMaRV) associated with future ICBM targeting studies.³

In the early 1980s, a Tactical Aerospace Assessment (TAA) Program was initiated under the auspices of the Air Force Systems Command's Advanced Weapons Concept Group to focus on ground targets which might be struck by a future Transatmospheric Vehicle (TAV). Rockwell (Downey), General Electric (Valley Forge), Martin-Marietta (Orlando), and Boeing (Seattle), among others, were involved in reentry designs and weaponization work to support a future TAV. Development of an atmospheric reentry capability was integral to striking earth and air breathing targets from a high-Mach TAV platform. By the late 1980s the Have Not Program, supported by Martin Marietta, was actively working on such a capability.⁴

The CAV Concept

The CAV concept is based on a multi-mission capable reentry system for global precision delivery or dispensing of payloads, cargo, or weapons through or from space. When launched from the Continental United States (CONUS) on any of several lift

vehicles (discussed later), the CAV's nominal time-to-target should normally be less than 90 minutes from launch. A synthesis of opinions suggests that the CAV should have a cross-range capability of more than 2,400 Nautical Miles (NM) for global engagement of multiple atmospheric or terrestrial targets from a pop-up maneuver or Low-Earth Orbit (LEO). The estimated weight of the CAV is 500 pounds, with a nominal cargo weight of 800 pounds, producing a gross weight of the CAV and payload of approximately 1,300 pounds. It must be robust enough to survive and operate in conditions produced by Mach 30 to sub-Mach speeds when transitioning from space to, and through, the Earth's atmosphere. The following major segments comprise the generic CAV reentry system concept of the Air Force Materiel Command (AFMC) Air Armament Center (AAC): 1) a rigid aeroshell segment with thermal protection system, 2) the propulsion segment, 3) the guidance and control segment, and 4) a payload release system.

- **Aeroshell and thermal protection**

The vehicle segment should: 1) be of such a shape and construction that it will withstand the atmospheric, space, and transitional environments imposed by repeated rapid acceleration and deceleration, and 2) be of such a size that it can be employed on rocket boosters and the future SOV for carrying miniature munitions and sensor cargoes.

- **Propulsion**

The propulsion segment for reentry positioning for dispersal or delivery of payloads, cargoes or weapons must be a safe, reliable, high-energy source capable of being stored for long periods of time.

- **Guidance and control**

The CAV guidance segment includes precision geo-location capability based on Global Positioning System (GPS) or inertial navigation systems, or a combination of the two, which will accurately and reliably position the CAV at a desired spot in three dimensional space for the release or dispensing of payload, cargo, or weapons. The evolution of CAV from earlier MaRV designs has led some to think of it as primarily an aerodynamically guided ballistic reentry vehicle, while others believe the CAV should be a state-of-the-art precision guided vehicle.

- **Payload Release**

A centerline ejection capability will dispense the cargo, payload, or weapon at the rear of the aeroshell so as to not perturb the flight of the CAV.

The AFMC AAC concept, shown in Figure 1, embodies the primary components of their CAV design.

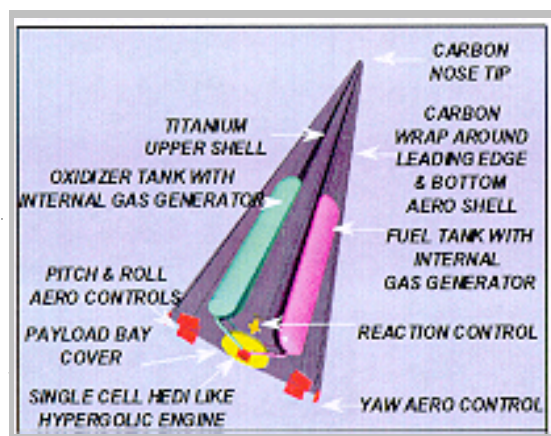


Figure 1. AFMC Air Armament Center CAV Concept.

The CAV delivery system has the following "common" characteristics: 1) it should be "common" by mating to many space launch platforms, such as rocket booster, space operations vehicle, or air launch systems; 2) it should be "common" by accommodating many cargoes, payloads, or weapons. Beyond these two "common" characteristics, the CAV concept evolved in several directions. Some advocates promote a steerable ballistic CAV much like the MaRV, AMaRV, or HpMaRV concepts of the past. Others foresee a fully powered and navigable CAV which possesses extensive orbital and atmospheric maneuverability. Whether the CAV should be an expendable delivery vehicle, used only once and destroyed as part of its mission profile; or a multiple-use system which is recovered, refurbished, and reused, has yet to be resolved.

Employing the CAV

The CAV employment profile consists of the following sequential segments: 1) launch of the CAV reentry aeroshell with its mated cargo or weapon into a ballistic trajectory or LEO, 2) atmosphere reentry, slowing, and guidance to a predetermined release point; 3) ejection of the cargo, payload, or weapon, and 4) recovery or destruction of the CAV. Differences in launch platform capability and mission scenario will produce either a pop-up flight profile or an orbital flight profile.

The pop-up profile, shown in Figure 2, is based on a rocket booster or SOV launch in which the CAV follows a sub-orbital (i.e., ballistic), exo-atmospheric

trajectory. The CAV would be boosted to a dispersal point, or LEO prior to dispersal, by an attached, expendable, upper stage or boost engine. This profile provides the CAV's greatest throw-weight capability, allowing heavier payload delivery to terrestrial targets.

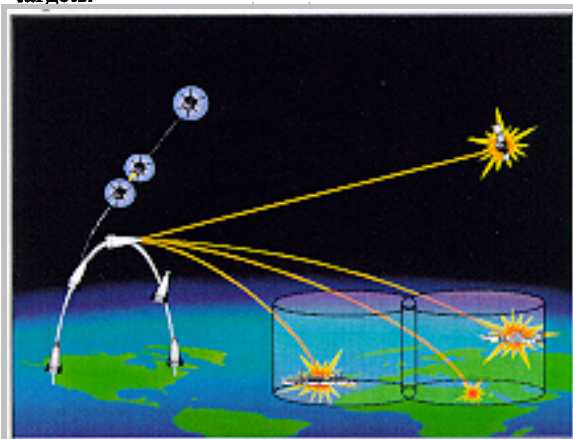


Figure 2. Suborbital "Pop-Up" Profile

The orbital profile, shown in Figure 3, occurs when the boost vehicle imparts to the CAV sufficient velocity to permit either a "once-around the Earth" trajectory or a sustainable LEO. Both of these modes of operation readily provide the capability for missions such as Force Enhancement (e.g., surveillance, reconnaissance, and intelligence gathering), and Force Application.

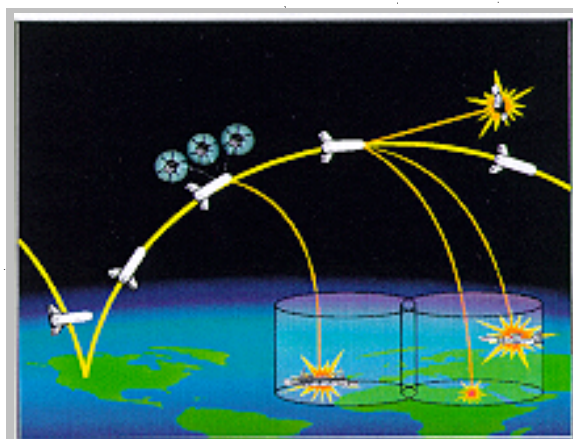


Figure 3. Orbital Profile

Launching Alternatives

Three types of launch or boost platforms have been identified which might be employed to loft the CAV into space. They are: 1) expendable ground-launched rocket, 2) expendable air-launched Pegasus like booster, and 3) Space Operations Vehicle. In the next decade, the most likely means of putting a CAV into space will be by either an expendable rocket or

an air-launched, Pegasus-like booster. Both of these launch systems involve proven technology and demonstrated performance. In the longer-term, the future Space Operations Vehicle (SOV) will become the launch platform of choice for the CAV.

Expendable Ground Launched Rocket

A number of commercial and military rockets are candidates for boosting a CAV into space. In the near-term, the expendable rocket appears to be the most likely platform available to boost the CAV based on the relative availability and reliability of rocket launch systems. Excess military boosters may be available for CAV employment as discussed in a previous paper published on the subject of Space Applications Of Surplus Intercontinental Ballistic Missiles by Matthew A. Bille.⁵



Figure 4. Notional Rocket Booster with CAV

Expendable Air Launched Rocket

Launch of the CAV on an Pegasus-like booster from beneath the wings of a large aircraft such as a B-52, a KC-10, or L-1011, at altitude is a near-term means of employing the CAV. Development of a CAV in the Have Not Program of the 1980s was allegedly based on an airborne air-launch capability.



Figure 5. Expendable Air Launched Rocket

Reusable Space Operations Vehicle (SOV)

The SOV, or the Aerospace Operations Vehicle (AOV) as the Air Force Scientific Advisory Board (SAB) *Report on A Space Roadmap for the 21st Century Aerospace Force* referred to it, will ultimately be the preferred reusable launch system for the CAV.⁶ The SOV, depicted in Figure 6, is expected to make access to space routine and should exhibit "aircraft-like" operations, implying that the SOV will exhibit operations tempo, maintenance, and reliability characteristics similar to the present mature air-breathing fleet. However, the SOV is not expected to be available until the second decade of the next century.



Figure 6. Notional SOV Launch with CAV

CAV Missions

The CAV will be capable of performing numerous delivery missions in both the Force Applications and the Force Enhancement mission areas. In carrying out these missions, the CAV will demonstrate a number of desired force characteristics deemed necessary by the Air Force to overcome and neutralize threats in the future strategic environment. Desired characteristics, as described in the Air Force Long Range Plan⁷ include:

- Ensure rapid response
- Emphasize extended reach
- Prepare for operations in/from the US
- Maintain high-tech edge
- Operate in non-traditional mediums
- Create hedges against uncertainty
- Strike globally with speed and precision

CAV Force Applications Capabilities

The global reach and promptness of the CAV suggests it could have its highest utility as a force applications tool in two cases: 1) the "halt phase" of combat; and 2) high value, highly defended, or time critical targets. A number of weapon concepts have been suggested for CAV deployment in support of these two.

The Air Armament Center of Air Force Materiel Command has developed a family of so-called Small Smart Munitions which might be deployed from a CAV in the halt phase of combat operations long before more conventional military forces reach the conflict. Blunting and containment of the forward advance of the enemy forces in the first tens of minutes of the conflict would have a strong psychological impact as well as the practical effect of making the battle area safer and more survivable for following friendly forces. The CAV-borne weapons capable of halting an attack in minutes were originally developed for the small bomb-bays of future air breathing fighter aircraft. These future weapons, as the adjectives "small" and "smart" imply, are smaller, being generally less than 250 pounds, smarter in their ability to find, identify, and precisely strike the specific targets they will attack, and are based on a modular weapons concept which addresses both fixed and mobile target types. Extremely accurate laser radar (LADAR) and computerized Weapons in the Small-Smart family include the Powered Low Cost Autonomous Attack System (LOCAAS), the Small Smart Bomb (SSB), the Dual Range Air-to-Air Munition (DRAAM), and the Brilliant Anti-Armor Technology (BAT) weapon. Low weapon weight means a CAV may be able to carry several weapons, thus multiplying the capability of an individual CAV mission.

Special high value, highly defended, or time critical targets may employ small, smart munitions too, but more specialized weapons may be required for special categories of targets such as hard and deeply buried targets, strategic relocatable targets, or biological and chemical targets. For example:

- A single Unitary Penetrator has been nominated as a CAV weapon to defeat deeply buried targets such as underground command and control complexes or buried production and storage facilities.
- A specialized Agent Defeat Weapon payload may be joined to a Unitary Penetrator to create a tailored weapon to defeat a specific type of biological or chemical weapon.
- Dispersed targets may require the CAV to dispense Highly Effective Area Attack Submunitions over a broader geographic area.
- A number of non-lethal weapon payloads, such as electronic jammers, pulse generators, or sticky foams, may be employed via CAV to deny, disrupt, disable, or destroy equipment or facilities.
- A UAV Hunter-Killer Payload tailored to a specific type of target might be deployed from a CAV to search for strategic relocatable targets.

CAV Force Enhancement Capabilities

CAV will be able to dispense, deploy, or deliver payloads which will support and enhance other military missions. For example, a variety of Command, Control, Communications, Computer, Intelligence, Surveillance and Reconnaissance (C4ISR) payloads could be delivered via CAV to the far side of the Earth and released as airborne sensors on small Unmanned Air Vehicles (UAV) to assess, report, and manage real-time activities in a specific local geographic area. Such capabilities might be employed for short-term localized intensive surveillance or as an interim gap-filler, preceding or augmenting other intelligence, surveillance, or reconnaissance systems. Northrop Grumman, for example, has designed a small UAV, called the Ferret, to carry reconnaissance sensors and possibly incorporate an ancillary strike capability, which might be used to support high-value or time-critical assessment, battle management, or interdiction for missions such as Agent Defeat Weapon or Strategic Relocatable Target.

CAV-deployed ground sensors

As an alternative to a UAV capability, a CAV might disperse mini-or micro-ground sensors in a localized geographic area which would report intelligence, surveillance, or reconnaissance information via satellite up-link back to military and civilian decision makers in the United States.

CAV-delivered parcels---a global UPS capability

The short time delivery of the CAV suggests a number of logistics and supply support scenarios requiring delivery of high-value, time critical, or urgent cargoes to geographically remote locations in near-real-time. Such missions may support conventional military forces or special operations anywhere in the world.

Final Disposition of the CAV

One of the unanswered questions in the CAV mission profile is whether the CAV should be reusable or expendable. If reusable, it would be necessary to fly the CAV to a recovery area or pluck it from the sky with a Halo-like system as was done with satellite payloads in the past. If expendable, the problem of destroying the CAV and the conclusion of its mission must be addressed. For if the CAV is not destroyed, it will be only a matter of time before our adversaries acquire a CAV and reverse-engineer it to threaten the US with their own CAV.

The Need for Global Projection Capability

In the past decade the convergence of an uncertain world security environment and an increasingly credible capability to field operational space systems has produced interest in projecting global presence and global power through and from space. The evolving post-Cold War world confronts civilian and military planners with a variety of competing political, social, economic and military issues which shape the possible futures of national defense and argue for a global force projection capability from the Continental United States. Even as national policy makers enunciate the desire for global engagement, global presence, and global power projection, they are confronted with the realities of constrained budgets, a shrinking U.S. military presence worldwide, and increasing threats to U.S. forces and interests around the globe.

Recent attacks on U.S. facilities and personnel, from Beirut and the Khobar Towers to the U.S. Embassies in Nairobi and Dar Es Salaam, indicate a rising vulnerability of forward based U.S. assets to violence. This trend toward a more hostile international environment, and its extrapolated implications, was captured in the December 1997 study *Transforming Defense*, when the National Defense Panel described a future world in which "allies might be coerced not to grant the United States access to their sovereign territory. Hostile forces might threaten punitive strikes (perhaps using weapons of mass destruction) against

nations considering an alliance with the United States."⁸

The Panel went on to say that: "... constraints on forward-basing (i.e. infrastructure outside the continental United States: ports, installations, prepositioned equipment, and airfields) and advanced technologies threaten to impede our access to regions."⁹ In the future, the challenge will be to conduct deep inland operations around the globe in the absence of access to forward bases. The Panel foresaw a future in which terrorism and technology conspire to make a space-strike capability (not associated with a forward presence) possibly the only power projection capability the United States can mount. The Panel essentially described the mission environment in which the CAV might be deployed as America's first line of force projection capability.

In a similar vein, a recent Rand Corporation study, *To Find, and Not to Yield*, also identified the trend to more capable adversaries in the future and greater difficulty in projecting and employing U.S. military power when it said: "Our potential enemies are not standing still--many already have lethal chemical and biological weapons and delivery vehicles, antiship mines and missiles, and capable air defenses, all of which can impede the deployment and employment of U.S. forces in hostile regions."¹⁰ The study went on to suggest that future U.S. forces would require "high-leverage enhancements" to counter such capabilities.

Military Futurists Look at Space

In addition to the National Defense Panel Study, several recent Air Force studies have looked at future environments and requirements in an effort to characterize future directions for the Air Force. Notable among these have been The United States Air Force, Air University *Spacecast 2020* study, the *Air Force 2025* study, the Air Force Scientific Advisory Board *New World Vistas* study, and the Air Force Scientific Advisory Board *A Space Roadmap for the 21st Century Aerospace Force*.

Spacecast 2020

The *Spacecast 2020* study, conducted in the 1993-94 academic year, focused on creating a vision of military space capabilities the United States would require in the year 2020 and beyond, and identifying the space systems concepts which showed the greatest potential for enhancing future space operations. The study developed a "most likely" future and a number of alternate futures. After a military space environment was developed for each scenario, a consensus world

view was forged that captured the most likely environment in which the United States would be operating. The consensus view became the basis for technology generation and assessment. Of the nineteen unique high-leverage space systems identified and subjected to operational analysis to rank the systems and their enabling technologies, the Refueled Transatmospheric Vehicle (TAV) was described as "clearly ahead of the rest" and the Orbital Maneuvering Vehicle was ranked near the top. Although the term CAV did not appear in the study, one can conclude that the CAV capability was inherent in the Orbital Maneuvering Vehicle description.¹¹

Air Force 2025

In the 1995-96 academic year Air University executed the *Air Force 2025* study, which focused on concepts, capabilities, and technologies the United States will require to remain the dominant air and space force in the future. Two volumes of the *Air Force 2025* study addressed space force application futures. The *Air Force 2025 Study, Space Operations: Through the Looking Glass, Global Area Strike* focused specifically on the opportunities inherent in future space-based strike systems. The study reviewed alternative technologies for global strike systems in the year 2025, and made recommendations for an optimum solution to provide prompt and flexible response capabilities. The study is significant for the number of alternative technologies and configurations considered.

Scenarios involving space-based platforms, intercontinental ballistic missiles (ICBM), and transatmospheric vehicles (TAV) were each considered in turn. The capability of ballistic missiles to carry non-nuclear payloads, in what could then be characterized as a conventional ballistic missile (CBM) role, was recognized as was a CAV-like capability to carry a variety of specialized munitions or submunitions. CAV-like capability was described in terms of stealthy, or very small, very agile, reentry vehicles that could complicate the problem of terminal defense. A second *Air Force 2025 Study, Hit'em Where It Hurts: Strategic Attack in 2025*, incorporated the space strike concepts developed in *Space Operations: Through the Looking Glass, Global Area Strike System* and opined that Transatmospheric Vehicles (TAV) will play significant roles in future strategic attack operations. It did not tackle the question of how such attack operations returning from space would survive reentry into, or through, the Earth's atmosphere.¹²

Scientific Advisory Board Studies

Several Air Force Scientific Advisory Board (SAB) studies have advocated the development of CAV-like space delivery systems. The Space Applications Volume of the 1995 SAB Study, *New World Vistas*, succinctly summed up the advantages and possibilities of striking from space when it said: "Future space systems will be well suited to project force against air, land and sea-based targets anywhere on earth. Precise delivery of munitions, directed energy or electronic warfare on virtually any target, heavily defended or not, within minutes or hours of tasking and with minimal risk to U.S. forces could have a decisive impact at all levels of conflict."¹³ Two of the SAB force projection recommendations for the Air Force were: 1) broaden the use of space to include direct force projection against surface, airborne, and space targets; and 2) develop space munitions capable of precision strike against surface and airborne targets.¹⁴

In the November 1998 *Report on A Space Roadmap for the 21st Century Aerospace Force*, the SAB was more specific concerning the need for a Munition Dispenser Reentry Vehicle. In comparing today's capability with tomorrow's promise, the SAB characterized the future Global Reach promise saying: "The aerospace force, with the right organization, training, and equipment, could deliver precisely calibrated effects, from taking a picture to dropping a precision munition, anywhere on earth, in less than an hour from the "go" order, with surprise and immunity to most defenses. Larger-scale deployments would be lighter, faster, and more effective, and the need to station forces in foreign theaters would be greatly reduced."¹⁵ The SAB Report employed two pictures (one of which is reproduced below) which it said "illustrate the great potential of integrated aerospace power." The figure below "sketches a scenario for a precision strike of a terrorist enclave or other time-critical target. It is based on a system capable of delivering precision-guided munitions at orbital speeds . . ." Such a system would permit destruction of the target in less than an hour from a National Command Authorities order with complete surprise, immunity to currently fielded active defenses, and a lower prospect of collateral damage. It could equally well conduct a photo reconnaissance mission to produce proof that a prohibited action was in progress."¹⁶

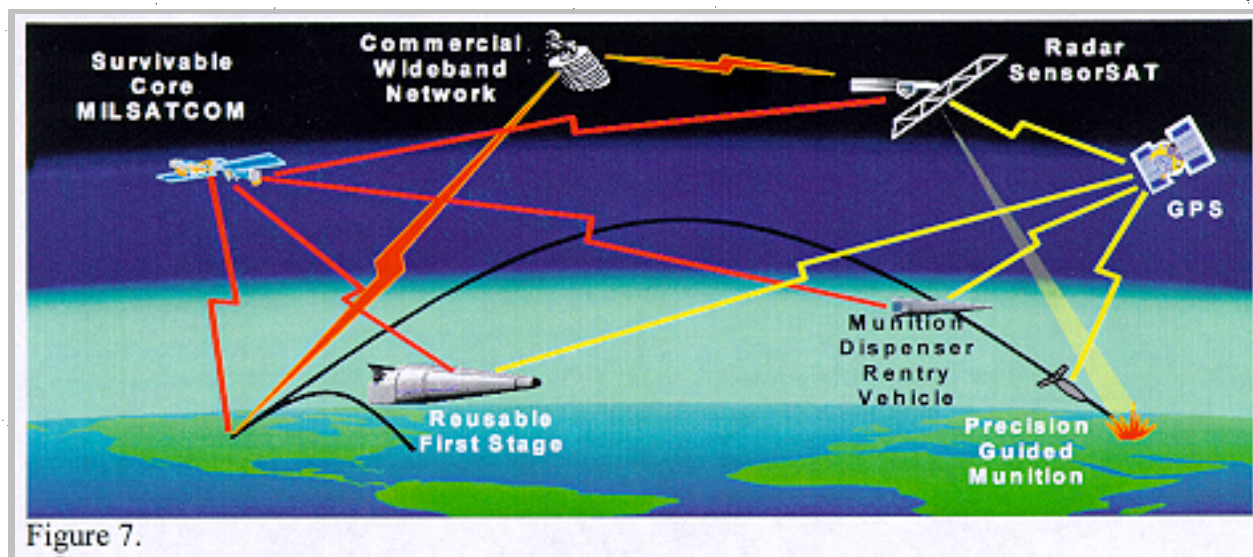


Figure 7.

Military Visions and Plans Supporting CAV

Over the past several years a cascading litany of visions and plans have described the need for future military forces to adapt to the changing world environment and adopt new, revolutionary, concepts for projecting military capabilities. From the most authoritative vision and planning documents one can identify the role of a CAV capability in the future. At the pinnacle of military visions stands the 1997 *Joint Vision 2010* of the Chairman of the Joint Chiefs of Staff, the overarching conceptual template which describes how the military will "channel the vitality and innovation of our people and leverage technological opportunities to achieve new level of effectiveness in joint warfighting." Recognizing that changes in the nature of our adversaries and their capabilities will require adaptation to a new and increasingly lethal battlespace, *Joint Vision 2010* states that, "long-range precision capability, combined with a wide range of delivery systems, is emerging as a key factor in future warfare." The Chairman's vision supports the importance of technological advances leading to improved precision and accuracy, enhanced stand-off capabilities, a wider range of delivery options, with a resulting economy of forces and higher tempo of operations.¹⁷

Joint Vision 2010 suggests that the nature of joint operations will change by the year 2010 and that instead of U.S. military forces relying on the physical massing of forces and subsequent sequential operations, other means will be employed to achieve massed effects. Two of the four new operational concepts described in *Joint Vision 2010* call for characteristics inherent in future space-strike systems such as CAV. *Dominant maneuver* calls for creation

of decisive speed and tempo and the creation of asymmetric advantages by attacking cross dimensionally, while *precision engagement* identifies the need for discriminate weapons strikes, and precise, all weather stand-off capability.

Global Engagement: A Vision for the 21st Century Air Force

Global Engagement: A Vision for the 21st Century Air Force describes how the Air Force will fight in the next century and builds the argument for future space-strike capabilities employing a reentry vehicle such as CAV. Recognizing that in the 21st Century it will be possible to "find, fix or track and target anything that moves on the surface of the earth, *Global Engagement* forecasts that "it then becomes possible to . . . hit an adversary's strategic centers of gravity directly . . . " and the ". . . ability to bring intense firepower to bear over global distances within hours to days . . .".¹⁸ *Global Engagement* recognizes the need for new means of leveraging technology to achieve asymmetric force advantages which strike directly at the enemy's ability to wage war. Foreseeing a future time when the Air Force will be forced to change the way it carries out its presence and power projection missions because of budgetary pressures, technological change, adversaries asymmetric strategies, and threats to forward bases, *Global Engagement* states that: "In the future, capabilities based in the continental United States will likely become the primary means for crisis response and power projection as long-range air and space-based assets increasingly fill requirements of the Global Attack core competency."¹⁹

CAV in the United States Space Command LRP

The *United States Space Command Long Range Plan* (LRP) Force Application Roadmap shows the CAV launched by a Conventional Ballistic Missile as the candidate space force application system to provide the key capabilities of *On Demand Force Applications*, applying force with decisive speed any place and anytime; *Flexible Force Application* against fixed, relocatable, or moving targets; and *Flexible Effects*, providing rapid, precise force projection featuring temporary and permanent effects in the 1998 to 2005 timeframe. The LRP Projected Assessment of Force Applications depicts the Conventional Ballistic Missile with CAV as becoming operational in the year 2004.²⁰

Air Force Basic Doctrine

The 1997 Air Force Basic Doctrine, which provides the officially sanctioned perspective, guidance, and warfighting principles that guide the employment of air and space forces in military operations, provides substantial support to space concepts such as CAV, advocating that: "We must continue to seek new, revolutionary, and imaginative ways to employ air and space power . . ."²¹ Air Force Basic Doctrine repeatedly stresses the global range and flexibility of air and space forces, emphasizing the characteristics inherent in space strike systems. "(Air and space) . . . forces have the capability to orchestrate maneuvers on a global scale and directly achieve strategic objectives."²² Likewise, ". . . security may be obtained by staying beyond the enemy's reach." Air and space forces have global reach and can strike at extended ranges.²³ Air and space forces can achieve surprise and shock based on their speed and global reach. "Space forces in particular hold the ultimate high ground, and as space forces advance and proliferate, they offer the potential for "permanent presence" over any part of the globe."²⁴

Recognizing the reality of post-Cold War downsizing and budget constraints, Air Force Basic Doctrine acknowledges that: "U.S. forces overseas have been reduced significantly, *while rapid power projection based in the continental United States (CONUS) has become the predominant military strategy.*"²⁵ (italics in the original)

From Plan to Reality

If a space-strike capability will be so desirable in the future, how do we get it? The process of moving from concept to reality in a major national defense program is long and convoluted. In today's world, the future

studies, concepts, and good ideas of the military futurists, scientific advisory boards, blue ribbon panels, and think tanks must be given shape and form in the long-range plans of the warfighter and the modernization planning process of the acquiring Service.

Based on the latest planning and requirements process, the AFSPC Integrated Planning Process describes a plan for each mission area. These mission area plans are brought together in the AFSPC Strategic Master Plan (SMP), which provides a roadmap for future needs and requirements. From SMP guidance and specific descriptions of needs flow, usually in the form of a mission need statement (MNS) or capstone requirements document (CRD), the activities which start the acquisition process. Next, an operational requirements document (ORD) defines in detail the requirement and a concept of operations (CONOPS) is created to describe the end-to-end operational solution scenario. That CAV is well along in this concept definition and requirement generation process as shown by the following CAV documentation.

Strategic Master Plan

The March 1998 Air Force Space Command (AFSPC) *Strategic Master Plan* (SMP) describes future Space Force Applications systems which might be either land-based or space-based. Ballistic missile land-based systems, as well as space-based systems, could provide prompt global attack capability against land, sea, and airborne targets. A Space Force Applications Roadmap depicts timelines and events for future space-strike systems, including a Conventional Ballistic Missile (CBM) with a modified non-nuclear reentry vehicle (MNNRV), and a CBM with a Common Aero Vehicle (CAV).²⁶

Mission Need Statements CAV Might Support

Several Air force Mission Need Statements identify force application requirements which could be satisfied by a CAV capability. For example, the Prompt Global Strike (PGS) Mission Need Statement (MNS), still in draft, is a clear call for a CAV capability. The latest draft of the PGS MNS describes the need for a global attack capability which can reach its destination in less than 120 minutes from launch.²⁷

The Strategic Relocatable Target (SRT) MNS describes the need to find, fix, track, and destroy road-mobile and rail-mobile missile launching systems carrying weapons of mass destruction (WMD), primarily nuclear weapons. Because SRT can move with some ease, the time from location /identification

until attack of the SRT must be very short. Additionally, SRT are high-value systems which are expected to be protected by heavy air defense. Both the relatively short time of flight of the CAV and its indifference to conventional terminal air defenses make it an excellent weapon system for striking SRT from space.²⁸

The purpose of the Hard and Deeply Buried Target Defeat Capability (HDBT/DC) MNS is described in its title. Relevant to the HDBT problem is: 1) facilities and bunkers will often be located deep inland or in-country and largely inaccessible to conventional tactical strike systems; and 2) such targets may be heavily protected by air defense systems. These characteristics make the HDBT targets excellent candidates for CAV delivery systems.²⁹

The Agent Defeat Weapon (ADW) MNS describes the need to destroy chemical and biological agents. The high-value nature of chemical and biological targets suggests: 1) they will often be protected in hardened or deeply buried production facilities or storage bunkers; 2) facilities and bunkers will often be located deep inland or in-country and largely inaccessible to conventional tactical strike systems; and 3) they will be heavily protected by air defense systems. These characteristics make the ADW targets excellent candidates for CAV delivery systems.³⁰

The draft Time Critical Target (TCT) Capstone Requirements Document (CRD) describes the need to attack fleeting and time-sensitive targets, similar to the SRT MNS above. Although the TCT CRD purports to address a theater need in near-real-time, the specific theater is not identified. Thus, the TCT requirement becomes a global requirement based on the non-specific theater to which one must respond. The promptness, deep in-land capability, and indifference to most defensive systems make the CAV a strong candidate to support the TCT CRD.³¹

Recent CAV Activities

Investigation and development of CAV concepts continues at Air Force Space Command, at the Air Force Research Laboratory, and within the Air Force Materiel Command's, Air Armament Center.

- The Air Armament Center continues to support CAV and small munitions concepts for eventual CAV development. The Air Armament Center is supporting a 1999 study to determine the characteristics and limitations of CAV launch via an expendable air launched rocket.

- The Air Force Space Command Directorate of Requirements is documenting the need for CAV and investigating strategies for creating a CAV program within the next Program Objective Memorandum cycle.
- The Air Force Space Command Directorate of Requirements and the Air Force Studies and Analysis Agency are planning a year 2000 Space Force Applications Military Utility/Worth Analysis which will examine CAV as an integral part of a space force application system of the future.
- The Air Force Studies and Analysis Agency (AFSAA) is investigating CAV weapons capabilities as part of a year-long (1999) multi-faceted Space Operations Vehicle study sponsored by the Air Force Space Command Directorate of Requirements.

Conclusion

The CAV is a concept which has captured the imagination of a variety of futurists, visionaries and military planners. In future decades it will provide the capability to accomplish a wide variety of missions from or through space. It gives new meaning to the terms global reach, global power projection, and global engagement.

While there remains much discussion on the details of CAV definition and design, there is general agreement on the mission and launch profiles of the Space-to-Earth delivery-dispenser capability. The means of boosting the CAV into space, its design characteristics, and the cargoes, payloads, or weapons it might carry and dispense are still open to discussion and will ultimately be decided based on the interplay of technological capabilities, mission needs, and cost. As the name common Aero Vehicle implies, the CAV is conceived as common in several ways: 1) common to several launch or boost platforms; and 2) common to many cargoes, payloads, or weapons. A CAV-like capability will be necessary to exploit the future capabilities of operating from or through space back to the Earth. It's not a question of if, but when, a CAV-like capability will be produced.

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References

- ¹ Air Force Museum Web Site:
www.wpafb.af.mil/museum/space_flight/
- ² *Boeing Ballistic Missile Offense Experience*, Boeing Phantom Works Briefing, 29 March 1999.
- ³ RV Development History, TRW Report, 1997.
- ⁴ Common Aero Vehicle (CAV) Briefing Slides provided by the Air Armament Center, Eglin AFB, FL on 25 August 1998.
- ⁵ Bille, Matthew A, *Space Applications Of Surplus Intercontinental Ballistic Missiles*, 1998.
- ⁶ *Report on A Space Roadmap for the 21st Century Aerospace Force*, Volume 1: Summary, November 1998.
- ⁷ *Air Force Long Range Plan*, Department of the Air Force, Deputy Chief of Staff for Plans and Programs, 1997, Page 5.
- ⁸ *Transforming Defense: National Security in the 21st Century*, Report of the National Defense Panel, December 1997, p. 12, 16.
- ⁹ *Ibid.*, p. 13, 21.
- ¹⁰ Ochmanek, David A., with Edward R. Harshberger, David E. Thaler, and Glenn A. Kent, *To Find, and Not To Yield*, Rand Corporation: Santa Monica CA, 1998, p. xvii.
- ⁵ *Spacecast 2020*, United States Air Force, Air University, 1994. Operational Analysis Volume.
- ⁶ *Air Force 2025, Hit'em Where It Hurts: Strategic Attack in 2025*, United States Air Force, Air University, 1994.
- ¹³ *New World Vistas, Air and Space Power for the 21st Century*, USAF Scientific Advisory Board, Space Applications Volume, 1995, p. ix.
- ¹⁴ *Ibid.*, p. ix, x.
- ¹⁵ *Report on A Space Roadmap for the 21st Century Aerospace Force*, Volume 1: Summary, November 1998, p. viii.
- ¹⁶ *Report on A Space Roadmap for the 21st Century Aerospace Force*, Volume 1: Summary, November 1998, pp. ix-x.
- ¹¹ *Joint Vision 2010*, Chairman of the Joint Chiefs of Staff, 1997, pp.1,11.
- ¹² *Global Engagement: A Vision for the 21st Century Air Force*, Department of the Air Force, 1997. P. 1.
- ¹⁹ *Ibid.* P. 11.
- ²⁰ *Ibid.*, P. 68.
- ²¹ *Air Force Basic Doctrine*, Air Force Doctrine Document 1, Secretary of the Air Force, September 1997, P. 40.
- ²² *Ibid.*, P. 37.
- ²³ *Ibid.*, P. 19.
- ²⁴ *Ibid.*, P. 25.
- ²⁵ *Ibid.*, P. 34.
- ²⁶ *Strategic Master Plan*, Air Force Space Command, March 1998, All detail from previous two paragraphs from pp. 39-42.
- ²⁷ *Draft Mission Need Statement (MNS) AFSPC 210-94 Prompt Global Strike (PGS)*, Air Force Space Command Directorate of Requirements, 15 July 1998.
- ²⁸ *Mission Need Statement (MNS) for Holding Strategic Relocatable Targets (SRT) at Risk*, United States Strategic Command 001-95/JROCM 070-97, 24 July 1997.
- ²⁹ *Hard and Deeply Buried Target/Defeat Capability (HDBT/DC) Mission Need Statement (MNS)*, CAF 317-92, 12 Oct 94.
- ³⁰ *Agent Defeat Weapon (ADW) Mission Need Statement (MNS)*, CAF 328-92, 5 October 1994.
- ³¹ *Draft Time Critical Target (TCT) Capstone Requirements Document (CRD)* CAF 313-97, 26 June 98.